

CHE 2201 – Physical Chemistry II

Phase Equilibria

Lec. 7. Solid Solutions

7.1. Introduction

Like liquid mixtures, two solids may exhibit complete miscibility or partial miscibility. Completely miscible solids form a stable solid solution, while partially miscible solids form a broken series of solid solutions.

To represent these systems, we will use the same temperature versus mole fraction diagrams. Since there are two phases (the liquid and the solid) with different concentrations, two curves have to be used to define the system. The temperature at which solids begin to deposit from the liquid solution is called the freezing point of the solution. The temperature at which the solid-solid solution begins to melt is called the melting point of the solution. The freezing point curve corresponds to the liquid phase and the melting point curve corresponds to the solid phase.

7.2. Completely Miscible Solids

Since the solids are miscible completely in each other, only one solid phase will be present in the system. If the components are completely miscible in the solid state, then they must be completely miscible in the liquid state as well. Hence, there will be only one liquid phase. As such, the appearance of the phase diagram is similar to that of completely miscible liquid pairs.

The addition of one component into the other component, may increase or decrease the freezing point of the other. For example, Cu and Ni are soluble in each other in all proportions in the solid state. The phase diagram for the Cu-Ni system is shown in Fig. 7.1.

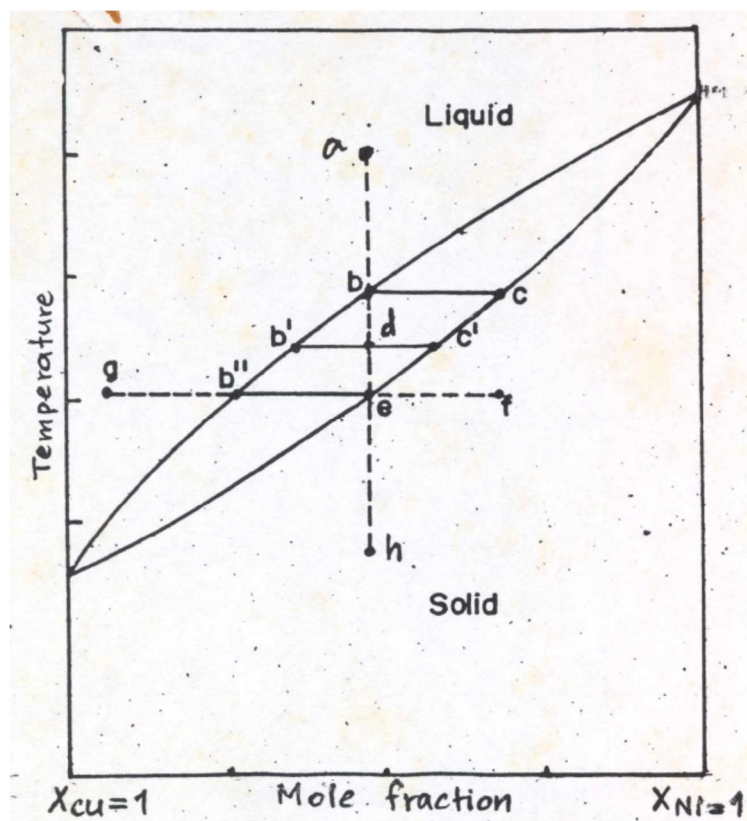


Fig. 7.1. Phase Diagram of the Cu-Ni System

The upper curve in the diagram is the freezing point curve (liquidus curve), and the lower curve is the melting point curve (solidus curve). The interpretation of the diagram is similar to that of liquid-vapour diagrams discussed in section 4.2. (See figures 4.2 and 4.4).

Binary systems which are known to form solid solutions over the entire range of composition, exhibit either a maximum or a minimum in the melting point. The melting point curve and the freezing point curves have an appearance similar to that of liquid-vapour curves in systems, which form azeotropes. This is same as an azeotrope that boils at a definite T . In case of solid mixtures having the composition at the maximum or minimum of the curve melts sharply like a pure substance.

Usually, the addition of another component lowers the melting point of the previous one. However, figure 7.1 illustrates a system in which the melting point of one component (Cu) is increased by the addition of another (Ni). The increase in the melting point can occur only if the solid in equilibrium with the liquid is not pure, but a solution.

Further, the phase diagram 7.1 shows that the mole fraction of Cu in solid solution is less than the mole fraction of Cu in the liquid solution ($X_{\text{Cu, Solid}} < X_{\text{Cu, Liquid}}$). Consequently, the melting point of Cu is **elevated**, whereas in the case of Ni, reverse takes place because the mole fraction of Ni in solid solution is greater than the mole fraction of Ni in the liquid solution ($X_{\text{Ni, Solid}} > X_{\text{Ni, Liquid}}$). Therefore, the melting point of Ni is decreased.

[Question: Explain the cooling behaviour indicated by the line abdeh, and the isothermal behaviour indicated by the line gb''ef.]

7.2.1. Fractional Crystallization of Solid Solutions

Fractional crystallization is a method used to separate pure solid substance from a solid mixture. For example, this technique can be used to separate pure Cu and/or Ni from a Cu-Ni mixture. This method of separation is very similar to the separation of pure liquids from liquids by fractional distillation (Refer to section 4.3).

7.3. Partially Miscible Solids

It is usual to find that two substances are neither completely miscible nor immiscible in the solid state, but each substance has a limited solubility in the other. The common type of the phase diagram is shown in Fig. 7.2.

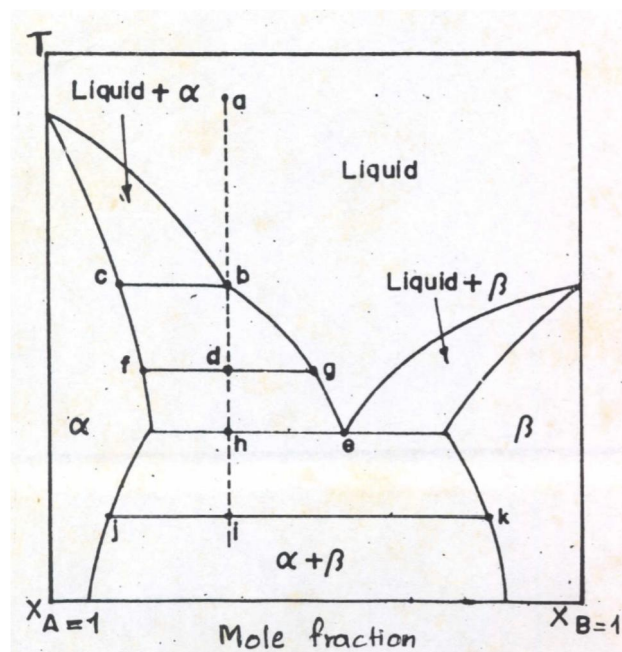


Fig. 7.2 Partial Miscibility in the Solid State

The points in the region α describe solid solutions of B in A, while the points in the region β describe solid solutions of A in B. The points in the region $(\alpha+\beta)$ describe states in which the two saturated solid solutions (two phases) α and β coexist in equilibrium.

Upon the cooling a system at point 'a', then at point 'b', solid solution α having the composition c appears. As the T decreases further, the compositions of solid and liquid shift; at point 'd', compositions of f and g are in equilibrium. At point 'h', the liquid has the eutectic composition e, and at this temperature solid β appears, which is in equilibrium with solid α and the liquid. On cooling to the point 'i', two solid solutions coexist. At this point, α (solid solution of B in A) will have the composition of j, and β (solid solution of A in B) will have the composition of k.

7.4. Alloys

Mixtures, compounds or solid solutions of two or more metals are known as alloys. The behaviour of alloys can be investigated by studying the freezing point curves of the metal mixture. According to this behaviour, the alloys can be divided into several groups. Each alloy may exhibit one or more of these characteristics.

(a). Components Separate Out in the Mixture

In this case, the freezing point curve is of the simple eutectic type (See Fig. 6.2). Since the metals tend to form solid solutions, this type of alloys are very rare.

E.g. Bismuth-Cadmium (Bi-Cd)

(b). Compound Formation

The two metals form one or more compounds with a **congruent** melting point (See Fig. 6.6).

E.g. Zinc-Magnesium (Zn-Mg)

(c). Solid Solution Formation

Please see Fig. 7.1.

E.g. Cu-Ni; Ag-Au

(d). Solid Solution with a Transition Point

Some metals form systems that increase the freezing point of the mixture by dissolving in the other component until a transition point is reached.

E.g. Ag-Zn

(e). Solid Solution with a Eutectic Point

E.g. Ag-Cu; Al-Au